



11051 - Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam Imaging SW				
	1		NIRSpec IFU Spectroscopy	(1) HVGC-1
	2		NIRCam Imaging	(1) HVGC-1

ABSTRACT

We propose deep NIRCam imaging and NIRSpec-IFU spectroscopy of HVGC-1, a hypervelocity globular cluster in the direction of M87. HVGC-1 has a velocity of -1035 km/s relative to the Sun, and 2300 km/s relative to M87. This significantly exceeds the escape velocity of M87 and that of the entire Virgo cluster, pointing to an exotic origin. The most plausible origin of HVGC-1 is a three-body ejection involving a supermassive black hole binary. In this case, the object is likely a stripped nuclear star cluster that hosts its own massive black hole. The distance, size, mass, and internal kinematics of HVGC-1 are uncertain. This program will measure its distance with 10% accuracy using the Tip of the Red Giant Branch method for resolved giants in the cluster's outskirts and resolve its spatial morphology, including a deep search for tidal tails. A distance measurement is key to constraining the time of HVGC-1's ejection and the velocity at which it was launched. IFU spectroscopy will map the cluster's internal velocity field and dispersion profile, measuring its dynamical mass and testing for the presence of a central black hole. Together, these observations will reveal the

structure and dynamical history of the first hypervelocity star cluster, which is likely the first known messenger from a supermassive black hole binary.

OBSERVING DESCRIPTION

We request deep NIRCam imaging and NIRSpec-IFU spectroscopy of HVGC-1, a hypervelocity globular cluster likely ejected from M87 in the Virgo Cluster.

NIRCam Imaging:

We will obtain NIRCam imaging in two short-wavelength (SW) filters, F090W and F150W, with simultaneous long-wavelength (LW) F322W2 exposures. Using two SW filters is essential to achieve the angular resolution and color baseline required to resolve the Tip of the Red Giant Branch (TRGB) at F090W = 26.7 (Vega) and measure stars up to two magnitudes fainter with high S/N. A total of 33.0 ks (16.5 ks per SW filter) will be executed with the FULLP full-frame subarray, no primary dithers, and a 12-point STANDARD subpixel dither pattern. This setup improves PSF sampling and bad-pixel/cosmic-ray mitigation while providing the full Module B field of view. The MEDIUMDEEP8 readout pattern with 9 groups per integration will be used. The simultaneous F322W2 image will provide complementary color information. The NIRCam field fully encompasses HVGC-1 and its surroundings, enabling measurements of its spatial profile, searches for faint tidal extensions, and the resolution of individual red giants to determine a precise TRGB distance.

NIRSpec-IFU Spectroscopy:

We will obtain integral-field spectroscopy using the NIRSpec IFU with the G140H grating and F100LP filter, providing a $3'' \times 3''$ field of view that fully encloses HVGC-1. A total of 35.2 ks of exposure time will be divided among four dither positions to improve spatial sampling and cosmic-ray rejection. We will use the NRSIRS2 readout pattern to suppress 1/f noise. Each observation consists of 6 integrations with 20 groups per integration (~1470 s each), short enough to minimize cosmic-ray impact as recommended by the NIRSpec team. The compact ($<1''$) source is suitable for the WATA target acquisition method. The spectra will yield spatially resolved velocity and dispersion maps to measure HVGC-1's internal kinematics.

Proposal 11051 - Targets - Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HVGC-1	RA: 12 30 54.6984 (187.7279100d) Dec: +12 40 58.62 (12.68295d) Equinox: J2000		
	Comments: Category=Stellar Cluster Description=[Globular star clusters] Extended=YES				

Proposal 11051 - Observation 1 - Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87

Observation	Proposal 11051, Observation 1											Tue Jun 02 16:00:27 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Observation 1) Warning (Form): Observers are responsible for checking that target acquisition is feasible.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	HVGC-1	RA: 12 30 54.6984 (187.7279100d) Dec: +12 40 58.62 (12.68295d) Equinox: J2000										
	Comments: Category=Stellar Cluster Description=[Globular star clusters] Extended=YES												
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947			
Template	HFF Readout Mode												
	false												
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140H/F100LP	NRSIRS2	20	6	false	true	NONE	4	24	35363.469		

Proposal 11051 - Observation 2 - Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87

Observation	Proposal 11051, Observation 2 Tue Jun 02 16:00:27 GMT 2026										
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(1)	HVGC-1	RA: 12 30 54.6984 (187.7279100d) Dec: +12 40 58.62 (12.68295d) Equinox: J2000								
Template	<i>Comments:</i> Category=Stellar Cluster Description=[Globular star clusters] Extended=YES										
	Module					Subarray					
Dithers	B										
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	NONE				STANDARD				12	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F322W2	MEDIUMDEEP8	9	1	12	12	16491.679		
	2	F150W	F322W2	MEDIUMDEEP8	9	1	12	12	16491.679		